



Aerodynamic Heating in the Gap Between a Missile Body and a Control Fin

Robert Alviani,* Devon Fano,† Jonathan Poggie,‡ and Gregory Blaisdell§
Purdue University, West Lafayette, Indiana 47907

<https://doi.org/10.2514/1.A35183>

Aerothermodynamic loading in the gap between a generic-missile-shaped body and a control fin was investigated through computations employing the Reynolds-averaged Navier–Stokes equations. The computed results for fully turbulent Mach 6 flow were compared with wind tunnel data obtained at the Arnold Engineering Development Center in 1979 experiments. For each case, the computational mesh encompassed the full geometry. Baseline computations of the missile body, in the absence of a fin, showed reasonable agreement with experimental data for heat transfer, surface pressure, and pitot pressure. The computations captured the order of magnitude increase in heat transfer over baseline levels with the presence of the fin and connecting cylinder. In agreement with the experiments, the computations predicted that minimizing the gap height minimized heat transfer levels. The maximum computed heat transfer levels on the missile surface occurred very close to the cylinder, a region that was inaccessible experimentally. In addition, heat transfer levels on the windward surface of the cylinder, which were not explored in the experimental study, significantly exceeded those on the missile surface for larger gap heights. The intense aerodynamic heating observed for this configuration highlights the high demands for thermal protection in the gap region under a control fin on a high-speed vehicle.

Nomenclature

C_f	=	skin friction coefficient
CFL	=	Courant–Friedrichs–Lewy number
D	=	cylinder diameter
g_h	=	gap height
h	=	heat transfer coefficient
M	=	Mach number
P_T	=	pitot pressure
$\dot{q}$	=	heat flux
R	=	specific gas constant
Re	=	Reynolds number
T_R	=	recovery temperature
T_W	=	wall temperature
X	=	distance from face of cylinder
x	=	distance from missile tip
α	=	angle of attack
Δs	=	wall-normal first cell grid spacing
δ	=	boundary-layer thickness
ϕ	=	angular coordinate (azimuthal angle)

Subscripts

BL	=	baseline value
0	=	stagnation
∞	=	freestream

I. Introduction

THE prediction of aerothermodynamic loading (i.e., surface pressure and heat transfer) under flight conditions is a fundamental

requirement for the design of air vehicles [1]. Both experimental and computational methods for characterizing aerothermodynamic loads have been developed and improved simultaneously over the last century. For low-speed aircraft, the prediction of important design parameters, e.g., coefficients of lift and drag, has become easier and more robust with the use of empirical and analytic methods. High-speed flight introduces complications, such as shock waves and expansion fans, which require specific experimental considerations, or the use of computational fluid dynamics, to accurately characterize. In addition, shock-wave/boundary-layer interactions can have a significant influence on vehicle performance by introducing flow separation, unsteadiness, and increased aerothermodynamic loading [2,3]. Wind tunnel experiments can capture these interactions but are costly, and measurements of important quantities can be difficult to obtain and are limited in scope. Because of these complexities, advancements in the field have relied heavily on computational fluid dynamics, which provides a characterization of the full flowfield and facilitates targeted measurement and experimental flexibility [4]. Computations, however, have varying degrees of accuracy and uncertainty depending on the physical model and numerical resolution. Thus, collaboration between experimental and computational studies allows for enhanced insight on specific flows, where both data sets can be used.

An important topic of high-speed flight vehicle research, which has received relatively little attention in the research community, is the effect of geometric imperfections on the aerothermodynamics of high-speed flow. Relevant geometric imperfections encompass any deviations from a smooth aerodynamic surface, such as roughness, steps, gaps, and cavities. The limited available research on gaps has universally shown intense aerodynamic heating in gap regions [5–8]. In general, the aerothermodynamic loading (i.e., surface pressure and heat flux) in gaps and cavities is positively correlated with increasing gap size, angle of attack, and Reynolds number [9,10]. For a wall-bounded flow with a gap, the loading is highly dependent on the incoming boundary-layer properties, with laminar flows producing significantly less heating than transitional or turbulent flows [11]. The studies mentioned all consider simple geometries, whereas complex gap regions such as the region under a control fin mounted on a missile body have been rarely studied. Thus, it is imperative to further investigate geometric imperfections and their resultant aerothermodynamic loading.

This paper presents a computational investigation of a generic missile configuration with a control fin attached by a cylindrical post. The primary focus of this research is to investigate aerodynamic heating in the missile-fin gap region around the post. Flow conditions

Presented as Paper 2020-0583 at the AIAA SciTech 2020 Forum, Orlando, FL, January 6–10, 2020; received 2 June 2021; revision received 19 November 2021; accepted for publication 30 November 2021; published online 27 December 2022. Copyright © 2021 by Robert Alviani, Devon Fano, Jonathan Poggie, and Gregory Blaisdell. Published by the American Institute of Aeronautics and Astronautics, Inc., with permission. All requests for copying and permission to reprint should be submitted to CCC at www.copyright.com; employ the eISSN 1533-6794 to initiate your request. See also AIAA Rights and Permissions www.aiaa.org/randp.

*Graduate Student, School of Aeronautics and Astronautics.

†Graduate Student, School of Aeronautics and Astronautics; currently Valcor Engineering, Springfield, New Jersey 07081.

‡Professor, School of Aeronautics and Astronautics. Associate Fellow AIAA.

were chosen to correspond to a previous experimental study [12]. Both a baseline missile, with no fin, and the missile-fin configuration were investigated. The purpose of this study was to evaluate the ability of a Reynolds-averaged Navier–Stokes model to replicate the experimental measurements and to provide further insight into the regions of the flow that were inaccessible experimentally. The structure of the paper is as follows. Section I provides an introduction to the research done for this paper. Section II outlines the methodology employed for this computational investigation. Section III provides the results for the baseline missile configuration. Section IV provides the results for the missile-fin configuration. Lastly, Sec. V discusses the results of this research and delivers concluding remarks.

II. Methodology

This computational investigation aimed to replicate the results of an experimental study carried out by Neumann and Hayes [12]. The Reynolds-averaged Navier–Stokes equations were employed in a three-dimensional computational model to simulate the experimental study. The angles of attack and gap heights included in the computed parametric study match those studied in the experiments. The initial set of computations corresponds with the baseline configuration, which was the missile model without the fin or connecting cylinder. Subsequent computations were then done on the missile-fin model for various gap heights.

A. Experimental Background

The Neumann–Hayes experiments [12] were carried out in the Arnold Engineering Development Center von Karman Facility Tunnel B (AEDC VKF Tunnel B) for Mach 6 freestream flow. The wind tunnel tests involved fully turbulent, high-Reynolds-number flow over an idealized missile model with, and without, an attached fin and connecting cylinder. Pitot pressure probes were employed for measuring the shock and boundary layers. Pressure transducers were used to obtain pressure data on the missile surface. Thermocouples were also inserted in order to calculate surface heat transfer on the

Table 1 Freestream flow conditions	
Parameter	Value
M_∞	5.95
$T_{0\infty}$	461.1 K
$P_{0\infty}$	1.724×10^6 Pa
u_∞	900.1 m/s
T_∞	57.06 K
P_∞	1149 Pa
ρ_∞	0.07017 kg/m^3
μ_∞	$3.753 \times 10^{-6} \text{ Pa} \cdot \text{s}$
Re_∞	$1.685 \times 10^7 \text{ 1/m}$

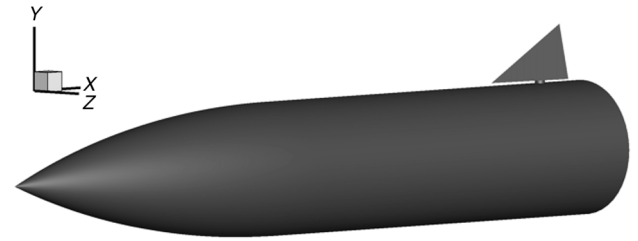


Fig. 2 Missile-fin configuration for a gap height of $g_h = 5.08$ mm. Coordinate system origin is at the nose of the missile body.

missile surface. The experimental freestream flow conditions, listed in Table 1, were identical for all cases considered here.

The missile model was a 1.27-m-long stainless steel ogive cylinder. The ogive section at the front extended for 0.572 m and the base diameter was 0.216 m. Initial experimental testing was done using the plain missile model, with no fin, designated here as the baseline configuration. Wind tunnel tests involved pitching the model at different angles of attack and taking measurements at various axial stations and angular coordinates. The model was only pitched downward, providing angles of attack ranging from 0 to -12° . The azimuthal angle ϕ was measured from 0 to 180° , with 0° corresponding to the windward, or top, centerline of the missile model. Figure 1 depicts this coordinate system. Two streamwise coordinates are employed in this paper, x and X . The distance x is measured from the missile tip, whereas X is measured upstream from the face of the cylinder: $X = x - (1.17 - D/2)$.

The missile-fin configuration, which includes the baseline missile, connecting cylinder, and fin, is illustrated in Fig. 2. The fin was a 0.152-m-long tetrahedron (four triangular sides), with a height of 86.75 mm and aft-thickness of 40 mm. The fin was situated toward the rear of the missile model. The tip of the fin was located at $x = 1.07$ m. A 15.9-mm-diam cylinder connected the fin to the missile model and was centered at $x = 1.17$ m. The gap between the missile and the fin would allow the fin to rotate as a control surface in a real flight vehicle. Experimental data were reported for gap heights g_h that ranged from 5.08 to 12.70 mm.

B. Mesh Generation

Two primary configurations were created, corresponding to the baseline missile model (no fin) and the missile-fin model. The commercial solid modeling software package Solidworks was used to reconstruct these geometries. The surface models were then transformed into computational meshes with use of the commercial grid generation program Pointwise. For each case, the computational mesh encompassed the full geometry; no symmetry boundary conditions were employed. To obtain higher computational accuracy, a structured grid approach was employed, where gridlines are aligned with no-slip boundaries and the leading-edge shock wave. Computational domains were sized to capture the entire leading-edge shock wave, for all angles of attack, as well as the wake region up to 0.5 m

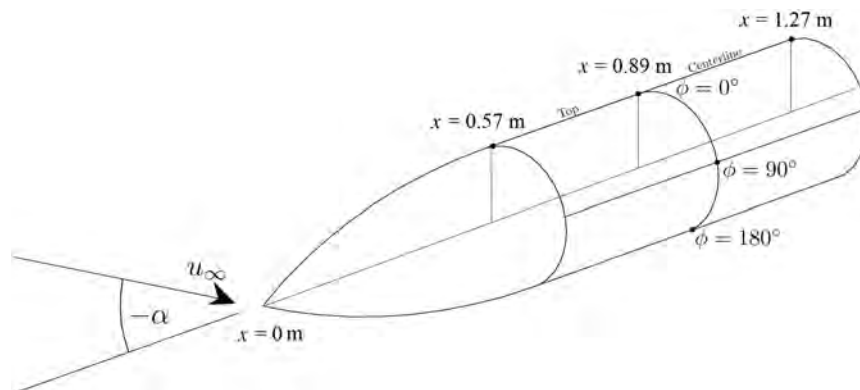


Fig. 1 Coordinate system for baseline missile configuration.

Table 2 Baseline mesh point counts			
Component	Coarse	Medium	Fine
A	200	400	800
B	18	25	45
C	100	150	300
D	48	71	141
E	6	7	10
F	20	35	65
G	60	80	140
R	25	45	75
T	9	17	32
Δs	8×10^{-6} m	4×10^{-6} m	2×10^{-6} m
Total points	1.4 million	5.6 million	48 million

behind the missile. The radius of curvature of the nose was 100 μ m. Component line distributions and total point counts for the coarse, medium, and fine baseline configuration meshes are provided in Table 2. Figure 3 shows the coarse baseline mesh with component lines labeled.

The missile-fin configuration domain comprised five structured blocks: the gap region, the region above the fin, the region adjacent to the fin, the missile, and the wake. The surface mesh on the fin bottom face was projected onto the missile surface to create the structured grid blocks in the gap region. Figure 4 shows the fin bottom face grid. The region near the fin tip was segmented into a four-sided diamond shape to accommodate a structured grid. Another structured grid wrapped around the cylinder and then connected with neighboring meshes. The Steger–Sorenson boundary control function was applied to smooth gridlines within and between adjacent blocks and to ensure orthogonality on grid boundaries [13]. The initial wall-normal spacing Δs on the missile surface and the cylinder surface for all meshes was 5×10^{-6} m and 2.5×10^{-6} m, respectively. In contrast to the baseline

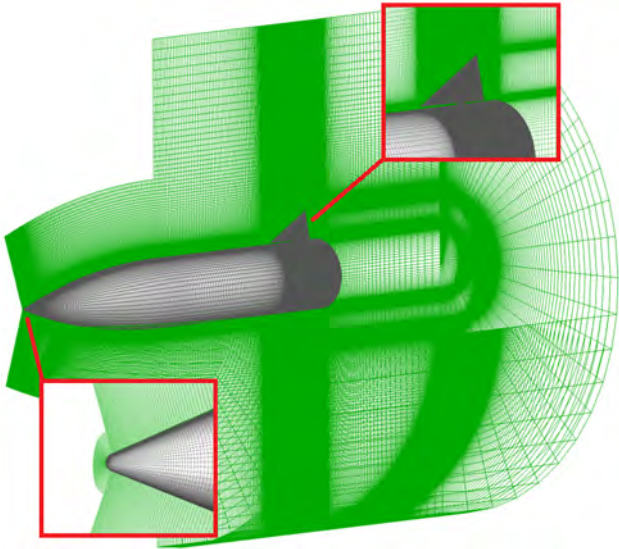


Fig. 5 Missile-fin mesh for $g_h = 5.08$ mm. Fin and nose regions highlighted and enlarged.

mesh, the missile-fin mesh was coarser on the leeward side than the windward side. The wake regions behind the fin and missile were meshed using surface extrusions. The entire missile-fin mesh, for a gap height $g_h = 5.08$ mm, is shown in Fig. 5. The number of points vertically along the cylinder ranged from 201 to 340, and the total points of the missile-fin meshes ranged from 78 million to 104 million, both depending on the gap height. No other component of the missile-fin mesh design varied with gap height, and the rest of the missile-fin mesh is identical to the baseline mesh.

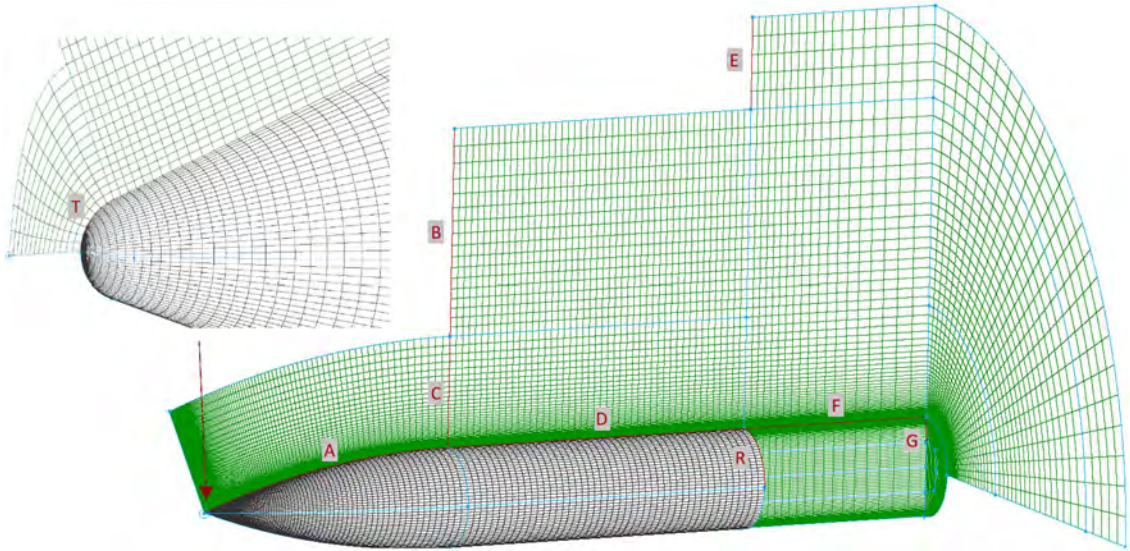


Fig. 3 Coarse baseline mesh with component lines labeled.

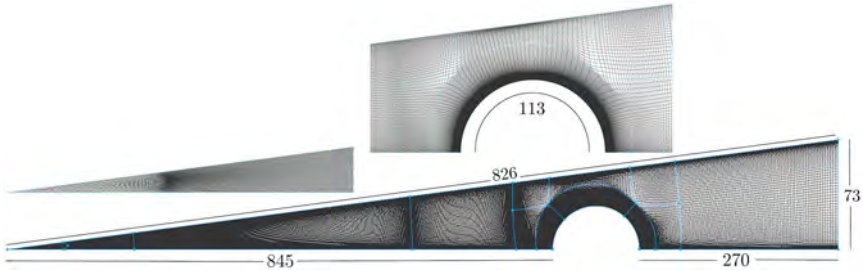


Fig. 4 Fin bottom face surface mesh with point counts labeled.

Table 3 Parametric study

Case	g_h , mm	α , deg
Baseline, coarse	—	0, -4, -8, -12
Baseline, medium	—	0, -4, -8, -12
Baseline, fine	—	0, -4, -8, -12
Missile-fin	5.08	0, -4, -8, -12
Missile-fin	7.62	0, -4, -8, -12
Missile-fin	10.16	0, -4, -8, -12
Missile-fin	12.70	0, -4, -8, -12

C. Computational Configuration

The Department of Defense software package CREATE-AV Kestrel was used to carry out all computations. Kestrel is a package of computational fluid dynamics solvers and tools that includes capabilities for prescribed body motion, propulsion, and multiphysics modeling [14]. The specific tool used for this work, Kestrel computational fluid dynamics (KCFD), is a finite-volume, unstructured, and cell-centered solver. The spatial accuracy was second order. The inviscid flux scheme was Harten-Lax-van Leer-Einfeldt (HLL++) and the viscous flux scheme was Limited Directed Difference (LDD+). Convective flux Jacobians were calculated with Van Leer's scheme. The flux limiter used was Kestrel+, which is based on the Barth-Jespersen scheme with an added shock wave sensor. The Menter Shear Stress Transport (SST) turbulence model was used for all the computations presented here. The solution was assumed to be steady state; therefore local pseudo-time-stepping was employed to accelerate convergence. The local time constraint was set to satisfy a Courant–Friedrichs–Lewy number $CFL = u\Delta t/\Delta x \leq 1$ at every cell. An isothermal cold-wall boundary condition was imposed at 300 K. The working gas was air and assumed to be thermally ($p = \rho RT$) and calorically ($e = C_v T$) perfect.

A total of 12 computations were done on the baseline configuration, and a total of 16 computations were done on the missile-fin configuration. For both configurations, a parametric study was done. For the baseline configuration, mesh refinement and angle of attack were varied. For the missile-fin configuration, gap height and angle of attack were varied. The details of each parameter variation are provided in Table 3. The gap heights and angles of attack were chosen to match the values used in the experiments.

III. Baseline Missile Results

The baseline configuration consisted of the missile body only, without the fin or cylinder. A grid resolution study was carried out, the overall flow structure was visualized, and heat transfer on the missile surface was analyzed and compared with experimental data. In addition, the baseline configuration established reference heat transfer levels for the missile-fin configuration heating analysis.

A. Grid Resolution Study

Coarse, medium, and fine meshes, outlined in Table 2, were created to compute the baseline configuration. Surface pressure, skin friction, and heat flux were analyzed to study grid independence. Figures 6–8 compare computed distributions of each property along the top centerline ($\phi = 0$ deg) of the missile for every angle of attack and mesh resolution. Values for the cylindrical section of the missile model are enlarged on the right-hand side in each figure. As expected, larger angles of attack were associated with greater levels of surface pressure, skin friction, and heat flux on the windward side. Distributions flattened near $x = 0.57$ m, where the ogive section of the missile model ended. The large spike in surface variables near $x = 1.27$ m can be attributed to the expansion fan at the aft-end of the missile body, which affects the subsonic portion of the boundary layer near the wall. This causes surface pressure to drop and heat flux and shear stress to rise. All properties slightly increased in value as mesh resolution increased. The average solution difference between the medium and fine results for surface pressure, skin friction, and heat flux on the flat part of the missile model was less than 2%. The results presented hereafter are from the fine mesh.

B. Overall Flow Structure

Contour plots of Mach number were generated to illustrate large-scale flow features and are provided in Fig. 9 for the fine mesh computations. As angle of attack increased in magnitude, the leading-edge shock wave on the windward side increased in strength, whereas the boundary layer on the windward side decreased in thickness. The leeward side flow experienced both a shock wave and an expansion region, and the detached flow at the end of the missile body increased in Mach number to levels above freestream. As angle of attack increased in magnitude, the wake region became increasingly asymmetric.

Surface pressure distributions were measured by Neumann and Hayes along the circumference of the missile model at $x = 1.07$ m for angles of attack of $\alpha = -4$ deg and $\alpha = -8$ deg. Figure 10 plots the computed surface pressure distributions compared with experimental distributions, where the angular coordinate system of Fig. 1 is used, noting that 0 deg is the windward (top) center. The computed distributions agreed well with the experimental distributions on the windward side, with average difference between computation and experiment less than 1%. However, there was discrepancy between the computed results and the experiment on the leeward side for $\alpha = -8$ deg, with a maximum difference between computation and experiment of 30%. This discrepancy may be due to boundary-layer separation and unsteady vortex formation, which the employed computational method would be unlikely to predict accurately.

Boundary-layer profiles were measured in the wind tunnel tests using pitot rakes. A row of closely spaced pitot tubes was placed near the missile model surface on the top centerline ($\phi = 0$ deg), at $x = 1.07$ m. Figure 11 plots the computed boundary-layer $P_T/P_{T\infty}$

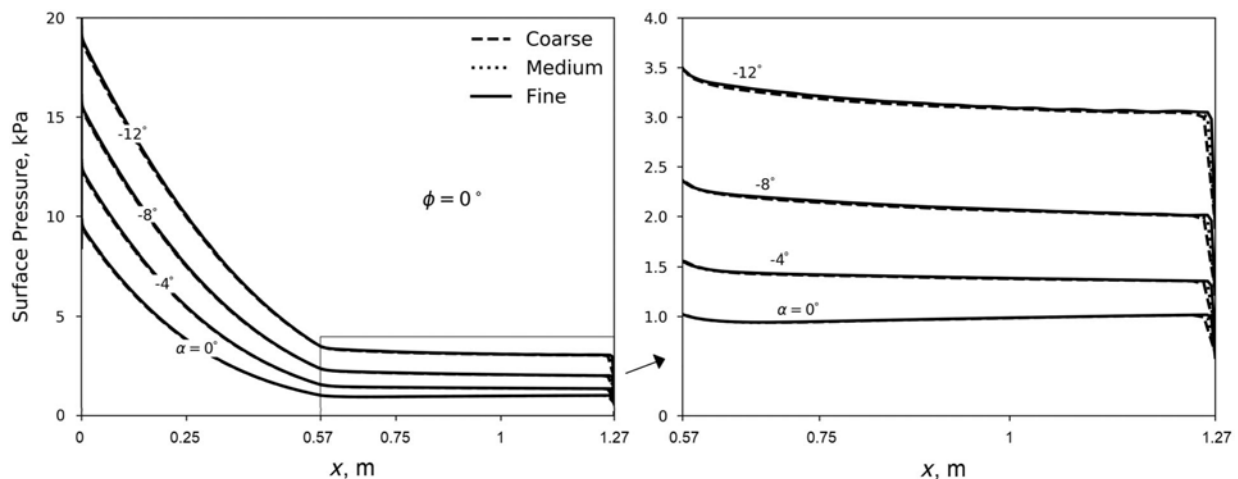


Fig. 6 Surface pressure P on top centerline ($\phi = 0$ deg). Cylindrical section of missile model enlarged.

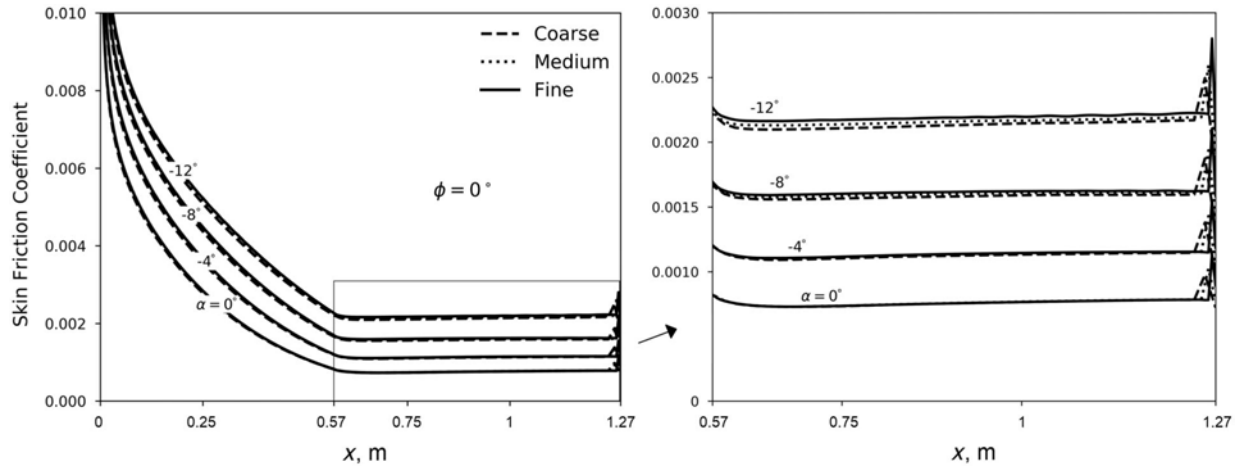


Fig. 7 Skin friction coefficient C_f on top centerline ($\phi = 0$ deg). Cylindrical section of missile model enlarged.

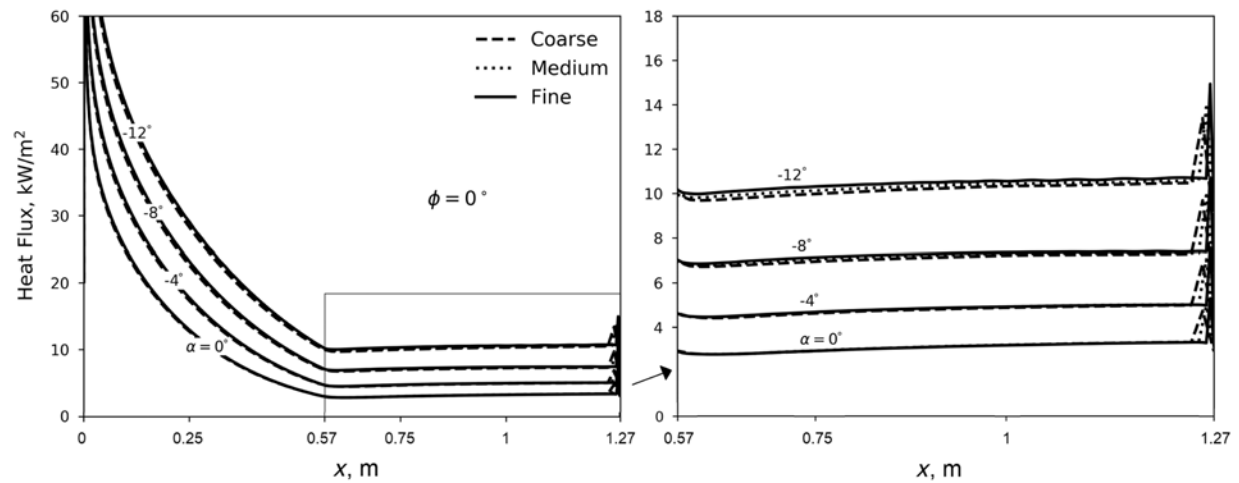


Fig. 8 Heat flux $\dot{q}$ on top centerline ($\phi = 0$ deg). Cylindrical section of missile model enlarged.

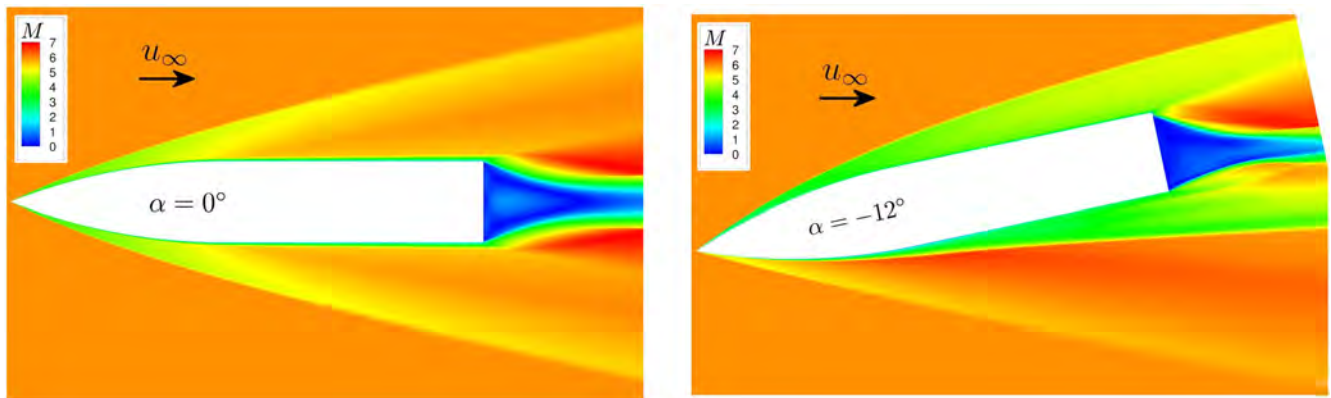


Fig. 9 Mach number contour plot of baseline computation for $\alpha = 0$ deg and $\alpha = -12$ deg. Centerplane slice from three-dimensional computation on fine mesh.

profiles alongside corresponding experimental data, with $P_{T\infty} = 52.9$ kPa. Values of boundary-layer thickness δ were determined by visually identifying knee locations on the plots. The computed boundary-layer thickness values agreed well with experimental boundary-layer thickness values. Also, the overall computed boundary-layer profiles for $\alpha = -8$ deg and $\alpha = -12$ deg agreed reasonably well with experimental profiles, with average difference between computation and experiment less than 5%. There was some discrepancy above and below the knee locations of the boundary-layer profiles for $\alpha = 0$ deg and $\alpha = -4$ deg. The exact cause of

this discrepancy is unknown; the experimental report does not report uncertainty for boundary-layer measurements [12]. Values for boundary-layer thickness are plotted against angle of attack and are provided in Fig. 12. A second-order polynomial curve fits the data well, which displays the $\delta(\alpha)$ relationship.

The experimental data included pressure measurements far above the surface to analyze the shock wave over the missile model at $x = 1.07$ m. Figure 13 plots the computed $P_T/P_{T\infty}$ against experimental data for these extended profiles. The computed values agreed very well with experimental data. The location of a change in slope

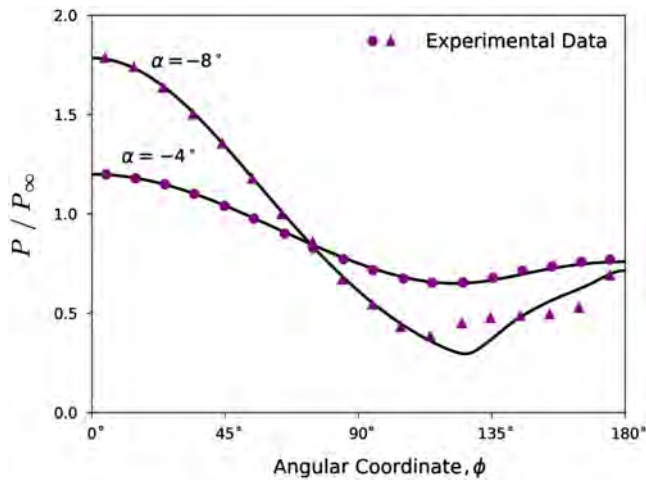


Fig. 10 Surface pressure P normalized by P_∞ along the missile model circumference at $x = 1.07$ m. Black lines are computed distributions (fine mesh).

above the boundary layer was accurately computed. There was discrepancy, however, regarding the location of the leading-edge shock front. The computations underpredicted the height of the shock wave. The fine mesh computation produced the closest match to the experiment.

C. Heat Transfer

Heat transfer rates measured in the experiments were given in the form of a heat transfer coefficient [15], defined as

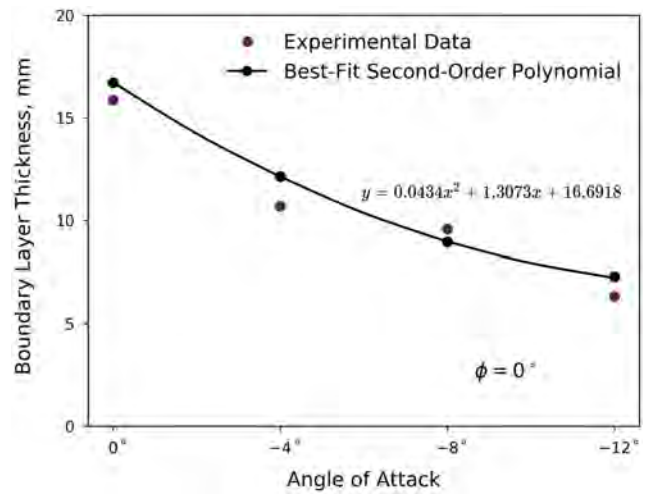


Fig. 12 Boundary-layer thickness on top centerline ($\phi = 0$ deg) at $x = 1.07$ m as a function of angle of attack. Black markers are computed values (fine mesh).

$$h = \frac{\dot{q}}{T_{\text{ref}} - T_w} \quad (1)$$

where $\dot{q}$ is heat flux per unit area in W/m^2 , T_w is the wall temperature in K, and T_{ref} is a chosen reference temperature in K. The heat transfer coefficient has units of $\text{W}/(\text{m}^2 \cdot \text{K})$. Factors that significantly affect the heat transfer coefficient include the boundary-layer state, local surface geometry, and the presence of separation or shock waves [16,17]. Heat transfer data were provided by the experimental

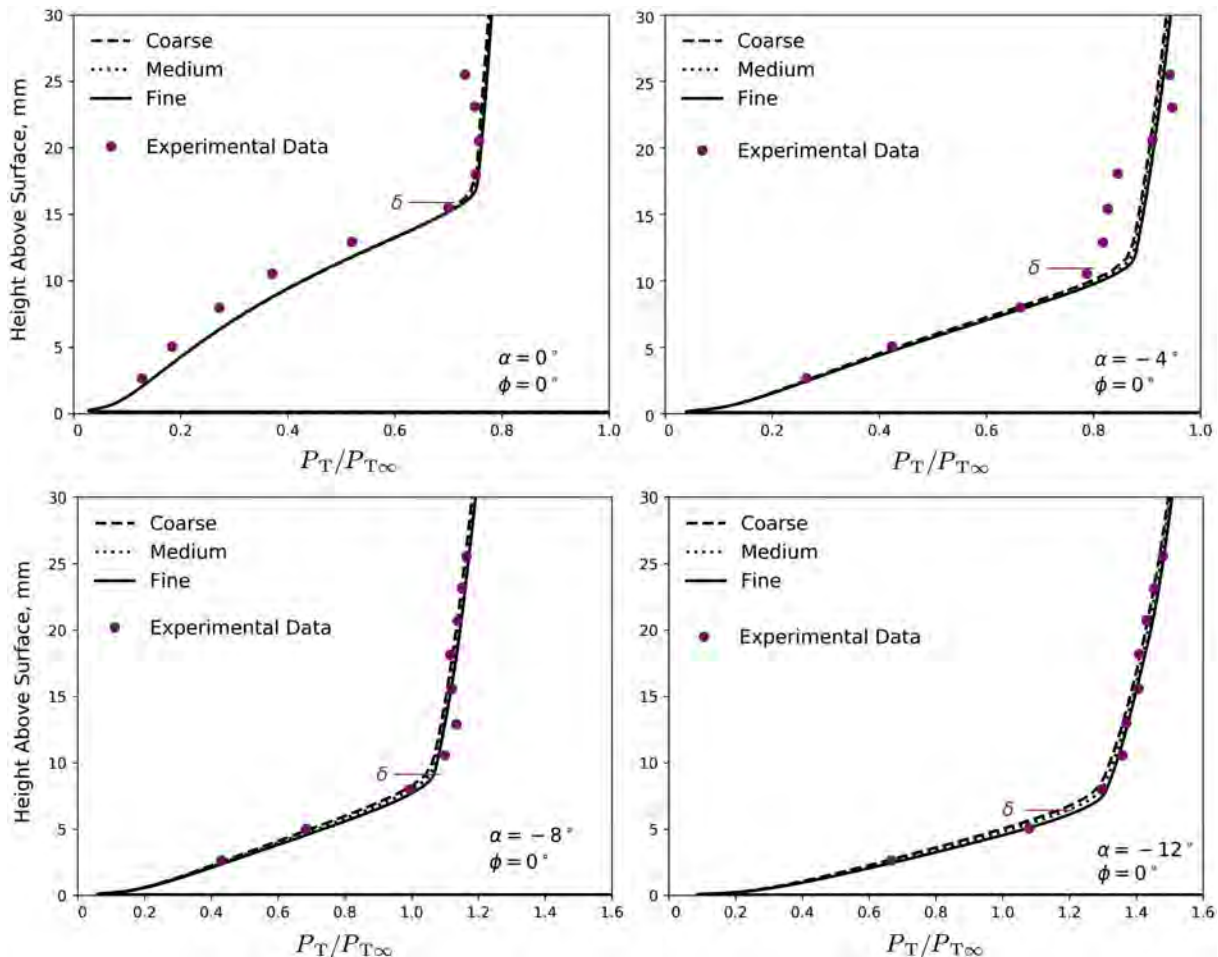


Fig. 11 Boundary-layer $P_T/P_{T\infty}$ profile at $x = 1.07$ m for angles of attack of 0, -4 , -8 , and -12 deg. Black lines are computed distributions (fine mesh).

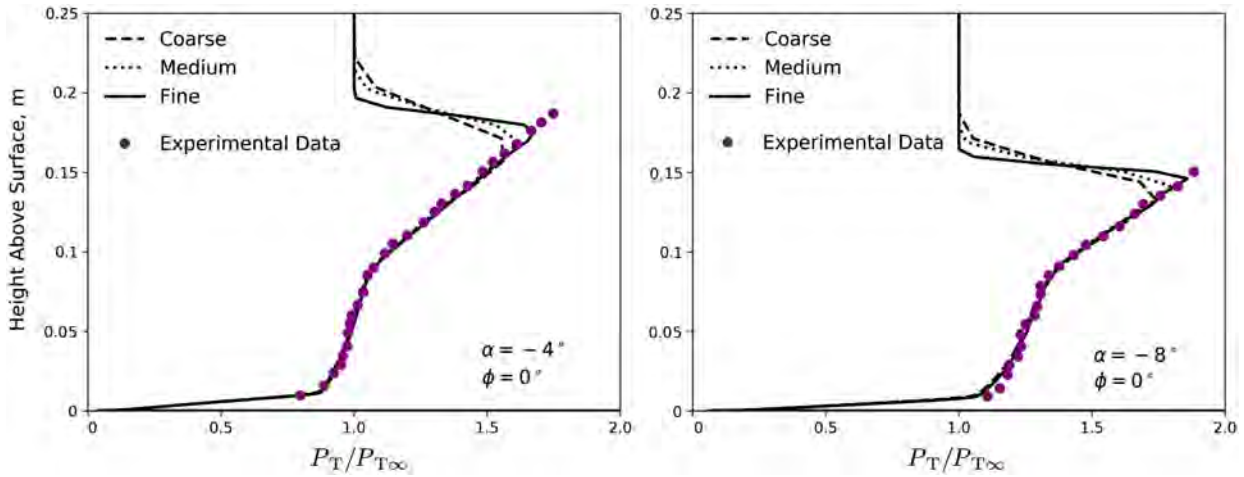


Fig. 13 Shock wave $P_T/P_{T\infty}$ profiles at $x = 1.07$ m for angles of attack of -4 and -8 deg. Black lines are computed distributions (fine mesh).

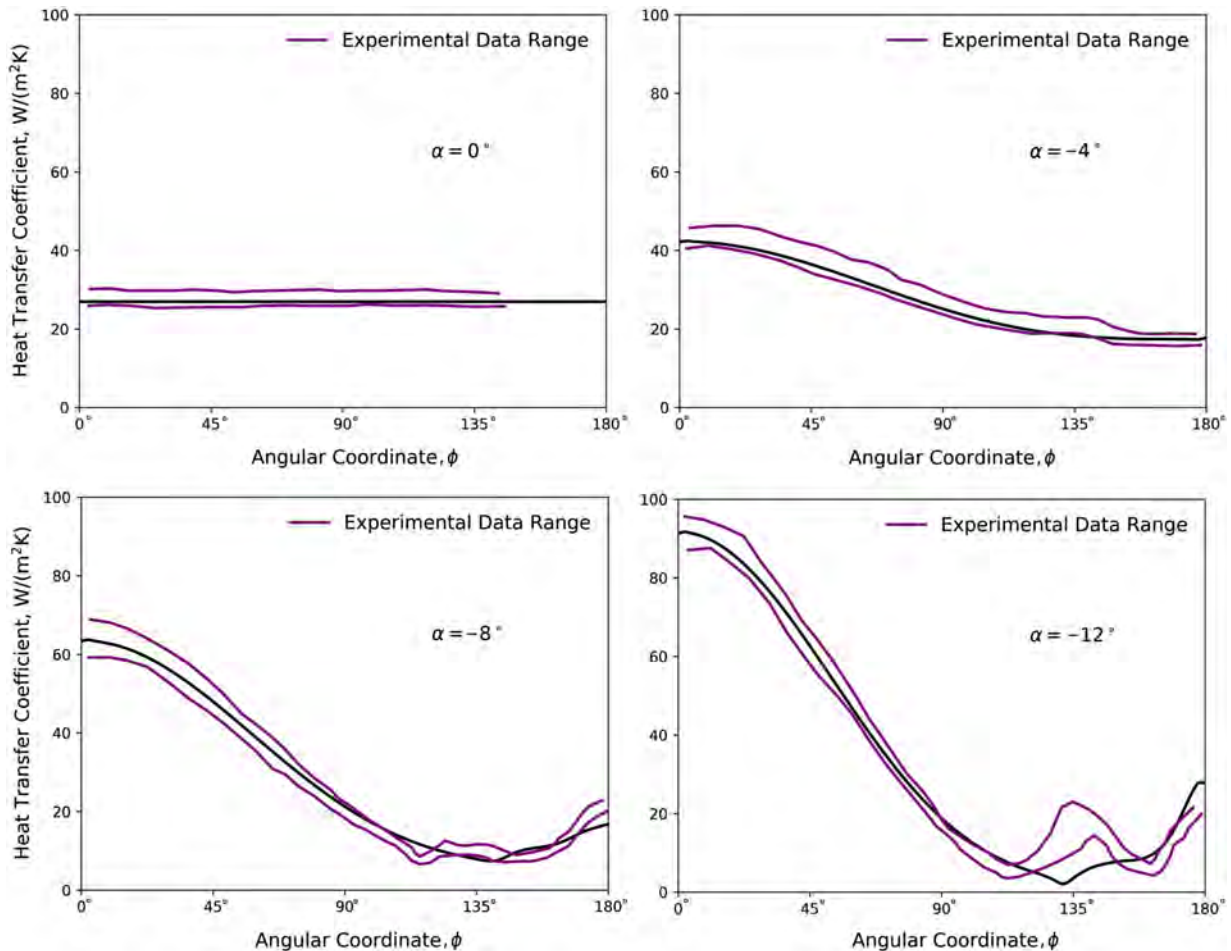


Fig. 14 Heat transfer coefficient along circumference of missile model at $x = 0.89$ m for angles of attack of 0 , -4 , -8 , and -12 deg. Black lines are computed distributions (fine mesh).

study along the circumference of the missile model at $x = 0.89$ m. There was considerable scatter in the experimental data; in the following figures, the data scatter is represented with upper and lower boundaries. Figure 14 displays the computed heat transfer coefficient distributions compared with the experimental measurements, with a reference temperature of $T_{\text{ref}} = 0.9T_{0\infty}$.

Computed heat transfer coefficient distributions agreed well with experimental distributions on the windward side, falling inside of the

experimental data range, but deviated from the measurements on the leeward side near $\phi = 135$ deg. This discrepancy on the leeward side was also present for calculations and measurements of surface pressure, shown in Fig. 10, which suggests the presence of separation. Levels of heat transfer and surface pressure were lower on the leeward side, however, so the maximum levels were accurately computed.

Figure 15 plots the heat transfer coefficient as a function of angle of attack on the top centerline ($\phi = 0$ deg), at $x = 0.89$ m, with

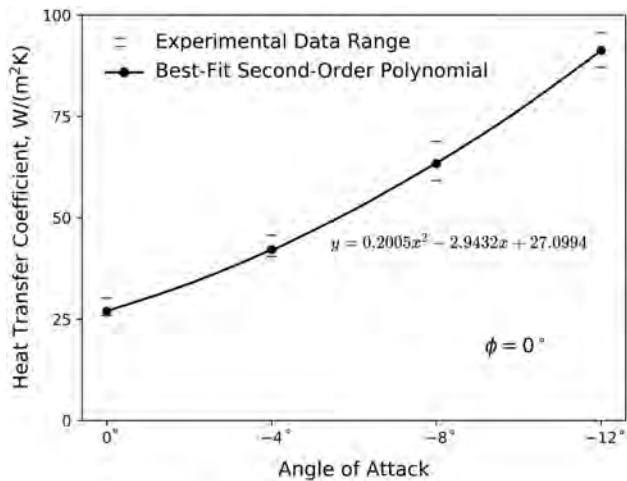


Fig. 15 Heat transfer coefficient at $x = 0.89$ m on top centerline ($\phi = 0$ deg) as a function of angle of attack. Black markers are computed values (fine mesh).

experimental measurement ranges included. Once again, a second-order polynomial is shown to illustrate the trend in the computed results.

IV. Missile-Fin Results

The missile-fin configuration involved the same missile body as the baseline configuration but with an added fin and connecting cylinder. The gap height between the missile and fin was varied to investigate its effects on heating in the gap region. The main region of interest for the missile-fin configuration was the top centerline ($\phi = 0$ deg) upstream of the cylinder in the gap region. All distributions plotted here were taken from that line. Figure 16 displays the fin and connecting cylinder geometries, with selected dimensions. The connecting cylinder diameter was 15.9 mm. The gap region extended $6.3D$ upstream, measured from the cylinder face to the fin tip. The fin's thickness increases linearly from the tip to end, as shown in Fig. 4, with an aft-thickness of 40 mm. Results presented here begin with the overall flow structure, which includes plots of Mach number, skin friction coefficient, and surface pressure. Discussion focuses on variations between cases in the parametric study. The heat transfer on the missile surface, scaled by baseline levels, is

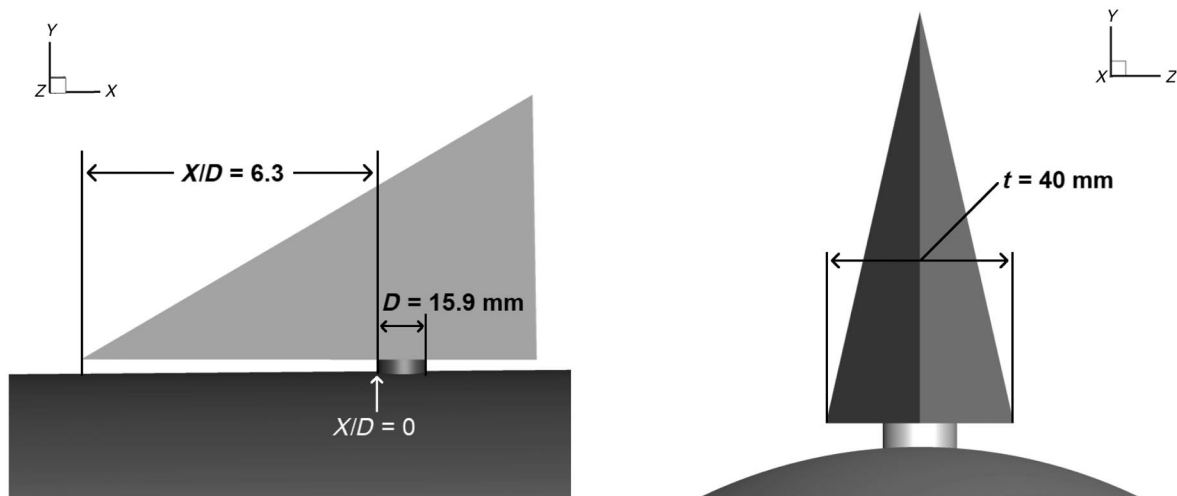


Fig. 16 Local fin and connecting cylinder geometry.

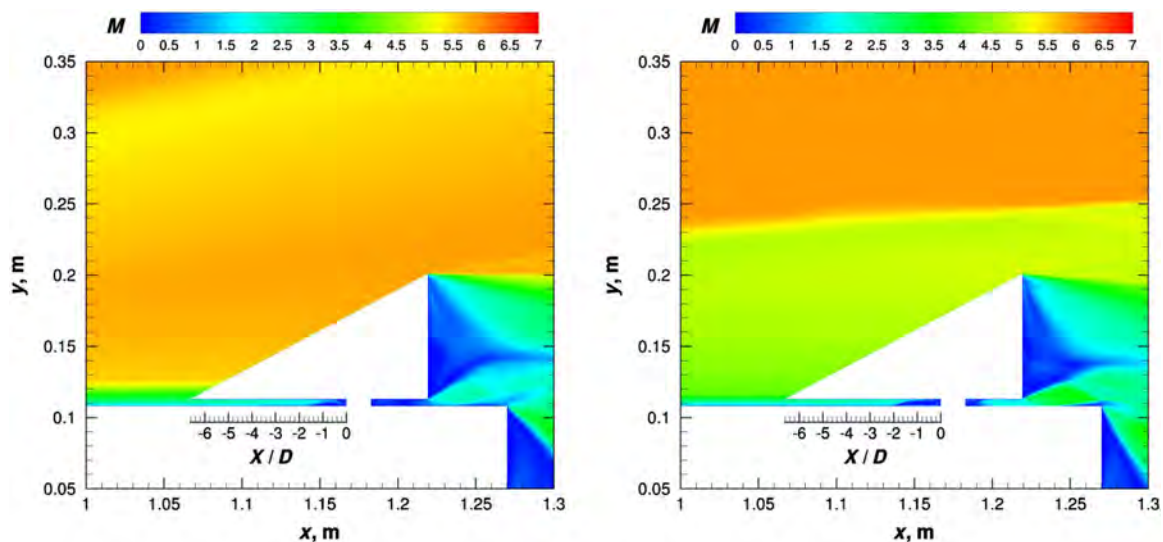


Fig. 17 Mach number contour plot of flow around fin region for a gap height of 5.08 mm and angles of attack of 0 and -12 deg. Centerplane slice from three-dimensional computation.

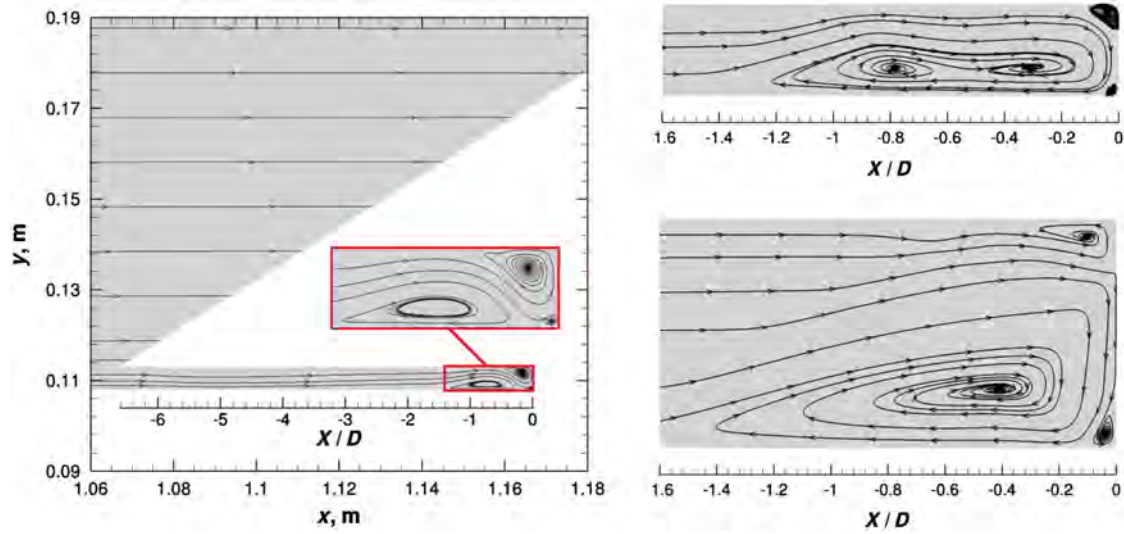


Fig. 18 Sectional streamlines of flow in gap region for a gap height of 5.08 mm and an angle of attack of 0 deg (left), for an angle of attack of -12 deg (top right), and for a gap height of 12.07 mm and an angle of attack of 0 deg (bottom right). Centerplane slice from three-dimensional computation. Near-cylinder separation region highlighted and enlarged.

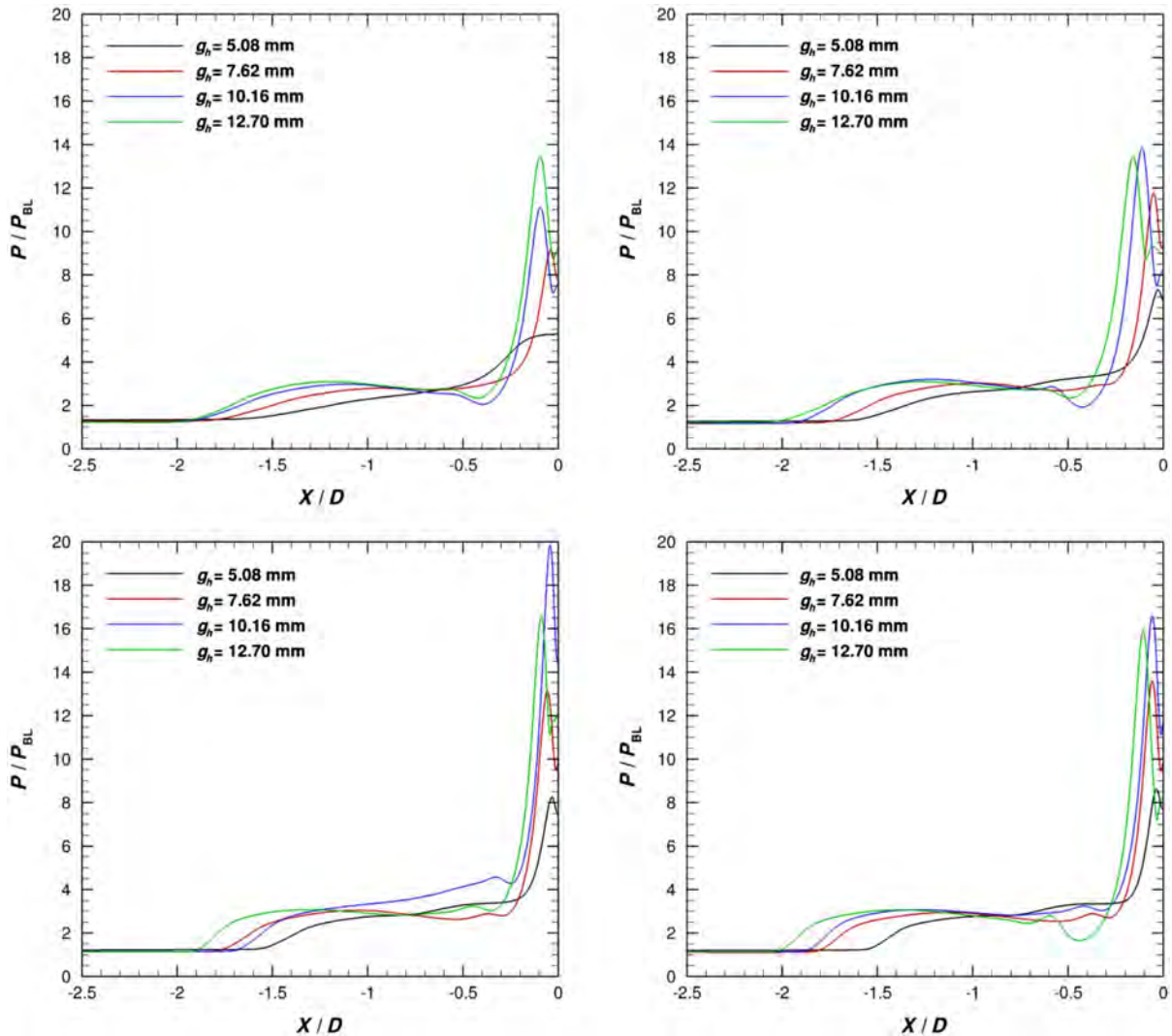


Fig. 19 Surface pressure on top centerline ($\phi = 0$ deg) in gap region normalized by baseline level. Angles of attack of 0, -4 , -8 , and -12 deg. Distance X measured from cylinder face.

compared with experimental data. Lastly, heat transfer on the cylinder surface is visualized and discussed.

A. Overall Flow Structure

A set of Mach contour plots were generated to illustrate general flow features around the fin. They are presented in Fig. 17 and are centerplane slices from the $g_h = 5.08$ mm computations for angles of attack of 0 and -12 deg. For the oblique shock wave above the top of the fin, downstream Mach number decreased with increasing angle of attack. The missile boundary layer remained attached until about $X/D = -2$. Flow separation was observed in the region between $X/D = -2$ and $X/D = 0$ for all cases and involved separation of the missile boundary layer, separation of the boundary layer on the bottom fin face, and at least two recirculation bubbles. The interaction creates a region of stagnation flow on the cylinder surface, where levels of maximum heat transfer occur. Figure 18 uses sectional streamlines to illustrate this separation for a gap height of $g_h = 5.08$ mm at $\alpha = 0$ deg and $\alpha = -12$ deg, and for a gap height of $g_h = 12.70$ mm at $\alpha = 0$ deg. As many as four separate vortices are seen in the gap region near the cylinder. For $g_h = 5.08$ mm at nonzero angles of attack, the primary recirculation region consists of two counter-rotating vortices. For greater gap heights, such as $g_h = 12.70$ mm, only one vortex is seen in the primary recirculation region; however, the two corner vortices are present in all computations.

Surface pressure distributions are displayed in Fig. 19. Values are normalized by the surface pressure on the top centerline

($\phi = 0$ deg), at $x = 1.07$ m, obtained from the fine mesh baseline computations for the corresponding angle of attack. Maximum levels of surface pressure occurred closer to the cylinder than $X/D = -0.3$, which corresponds to the nearest experimental probe to the cylinder. For the majority of cases, surface pressure upstream of $X/D = -0.5$ was less than or equal to four times the baseline level. Downstream of $X/D = -0.3$, surface pressure peaked at 20 times the baseline level for the case where $g_h = 10.16$ mm and $\alpha = -8$ deg. The behavior shown here is a result of the flow separation that occurs slightly ahead of the cylinder. Just downstream of the peak pressure rise, the pressure drops in the small corner vortex that can be seen in Fig. 18. For every angle of attack, maximum normalized levels of surface pressure increased with increasing gap height.

Skin friction distributions are plotted in Fig. 20. The distributions generally became negative, which indicates boundary-layer separation and the start of the recirculation region, near $X/D = -2$. The separation locations for larger gap heights were, for the most part, farther upstream than for those of smaller gap heights. Angle of attack seems to have had little effect on location of separation. Very close to the cylinder, skin friction distributions returned to positive values, indicating reattachment and the formation of the small corner vortex seen in Fig. 18. The locations of reattachment generally coincided with the locations of maximum heat transfer.

B. Heat Transfer

Plotted values of heat transfer presented in the section are normalized by the heat transfer coefficient values on the top centerline

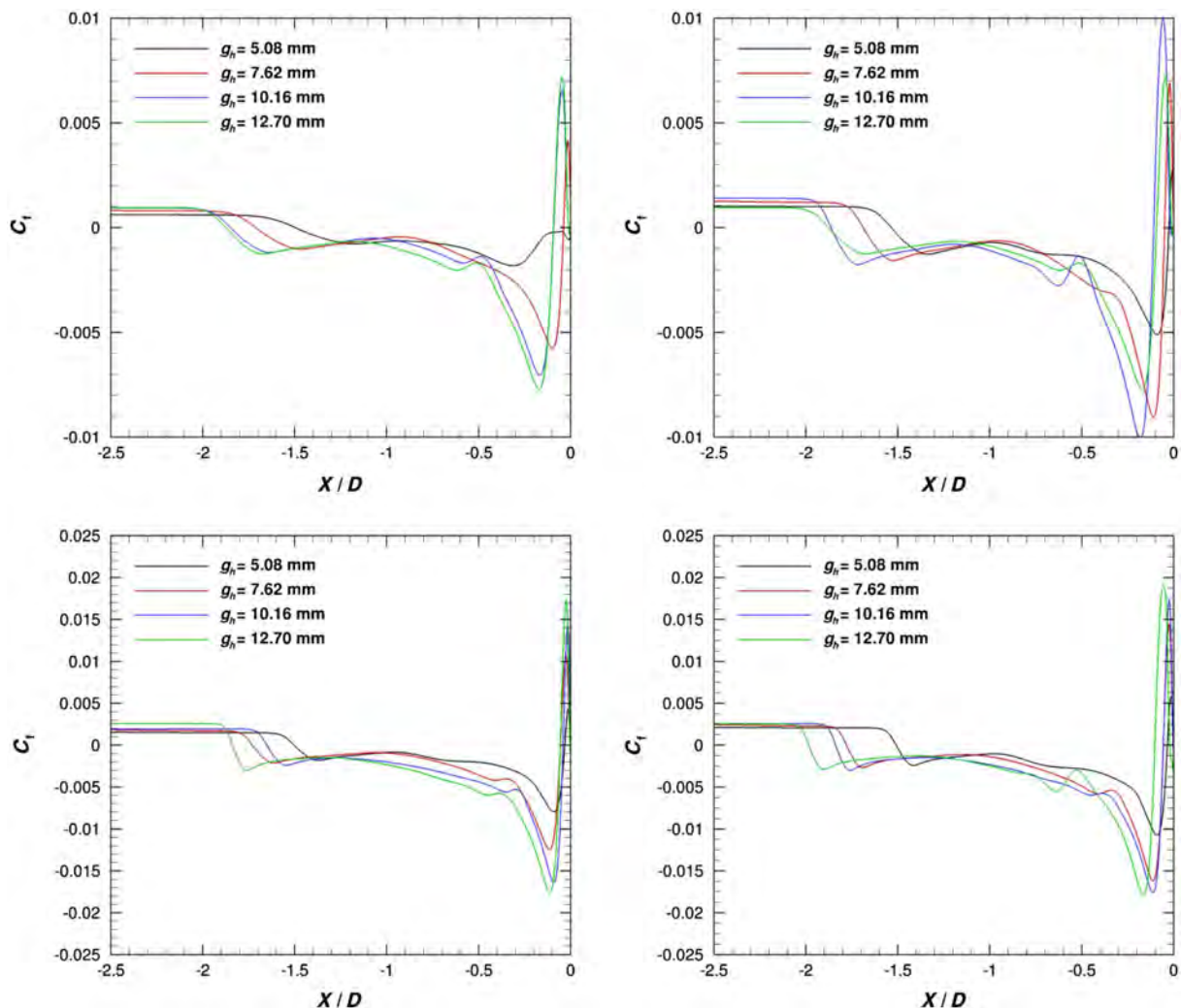


Fig. 20 Skin friction coefficient on top centerline ($\phi = 0$ deg) in gap region for angles of attack of 0, -4 , -8 , and -12 deg. Distance X measured from cylinder face.

($\phi = 0$ deg), at $x = 1.07$ m, obtained from the fine mesh baseline computations. These baseline levels were 28.1, 43.2, 64.1, and 92.1 W/(m² · K) for angles of attack of 0, -4, -8, and -12 deg, respectively.

Computed heat transfer coefficient distributions along the top centerline ($\phi = 0$ deg), for cases where $g_h = 7.62$ mm, are plotted and compared with experimental distributions in Fig. 21. All computed heat transfer coefficient values were defined using Eq. (1) with a reference temperature of $0.9T_{0\infty}$. Upstream of $X/D = -2$, computed heat transfer coefficient distributions agreed well with experimental distributions, with average difference between computation and experiment less than 10%. The computed distributions began to rise slightly downstream of $X/D = -2$, where the experimental heat transfer coefficients began to rise. At $X/D = -0.3$, all computed heat transfer levels were below experimental levels. The computed heat transfer coefficient distributions reached maximum levels very close to the cylinder. Maxima were computed to be much higher than experimental values measured at $X/D = -0.3$, which was the nearest experimental probe distance to the cylinder. The computed maximum levels occurred closer to the cylinder, around $X/D = -0.1$, and ranged between 20 and 22 times baseline levels.

Plots of the computed heat transfer coefficient distributions for every case are presented in Fig. 22. In addition, Fig. 23 provides the maximum heat transfer coefficient on the missile surface in the gap

region for every gap height as a function of angle of attack. For almost every angle of attack, maximum heat transfer coefficient levels increased with increasing gap height. Maximum heat transfer coefficient levels did not follow a monotonic trend with angle of attack.

The maximum computed heat transfer levels on the missile surface occurred very close to the cylinder, a region that was inaccessible experimentally. The maximum experimental values were never above 11 times the baseline level, always occurring at the closest probe to the cylinder at $X/D = -0.3$. As shown in Fig. 22, the computed maximum heating occurred significantly closer to the cylinder and reached much higher levels than were observable experimentally.

One aspect of the aerodynamic heating that was not explored in the experimental study is the heating on the cylinder surface, which is shown to far exceed the heating on the missile surface. Heating contour plots for the missile and cylinder surfaces in the gap region are presented in Fig. 24 for various gap heights and angles of attack. The band of maximum heat transfer on the missile surface near $X/D = -0.1$ is identifiable. The flow near the cylinder exhibits two regions of separation with a stagnation point between the two, which creates exceptionally high rates of heat transfer on the cylinder surface. This is shown to be as high as 80 times the baseline levels for the largest gap height $g_h = 12.70$ mm at -8 deg angle of attack.

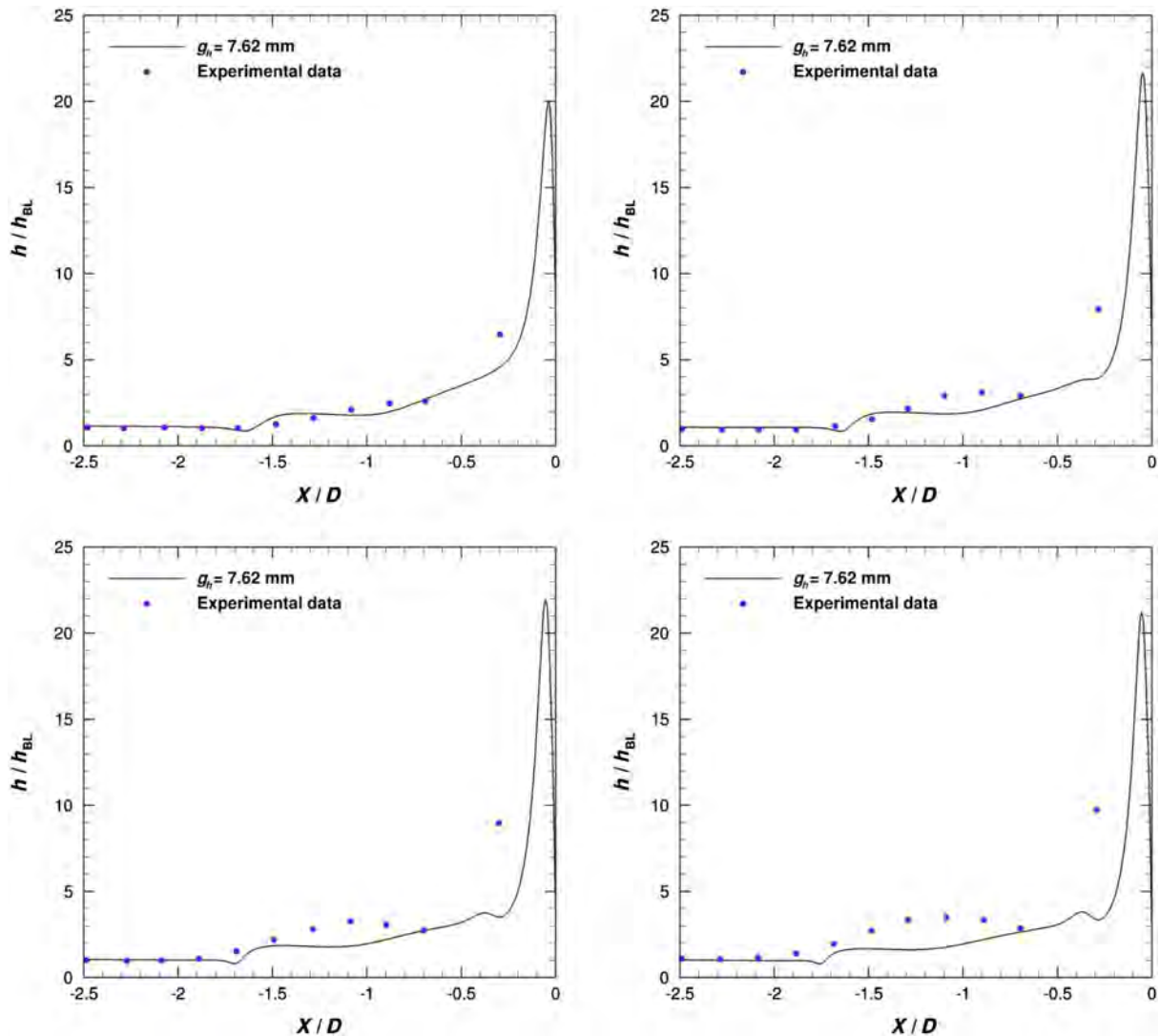


Fig. 21 Heat transfer coefficient normalized by baseline level on top centerline ($\phi = 0$ deg) in gap region. Gap height of 7.62 mm and angles of attack of 0, -4, -8, and -12 deg. Black lines are computed distributions. Distance X measured from cylinder face.

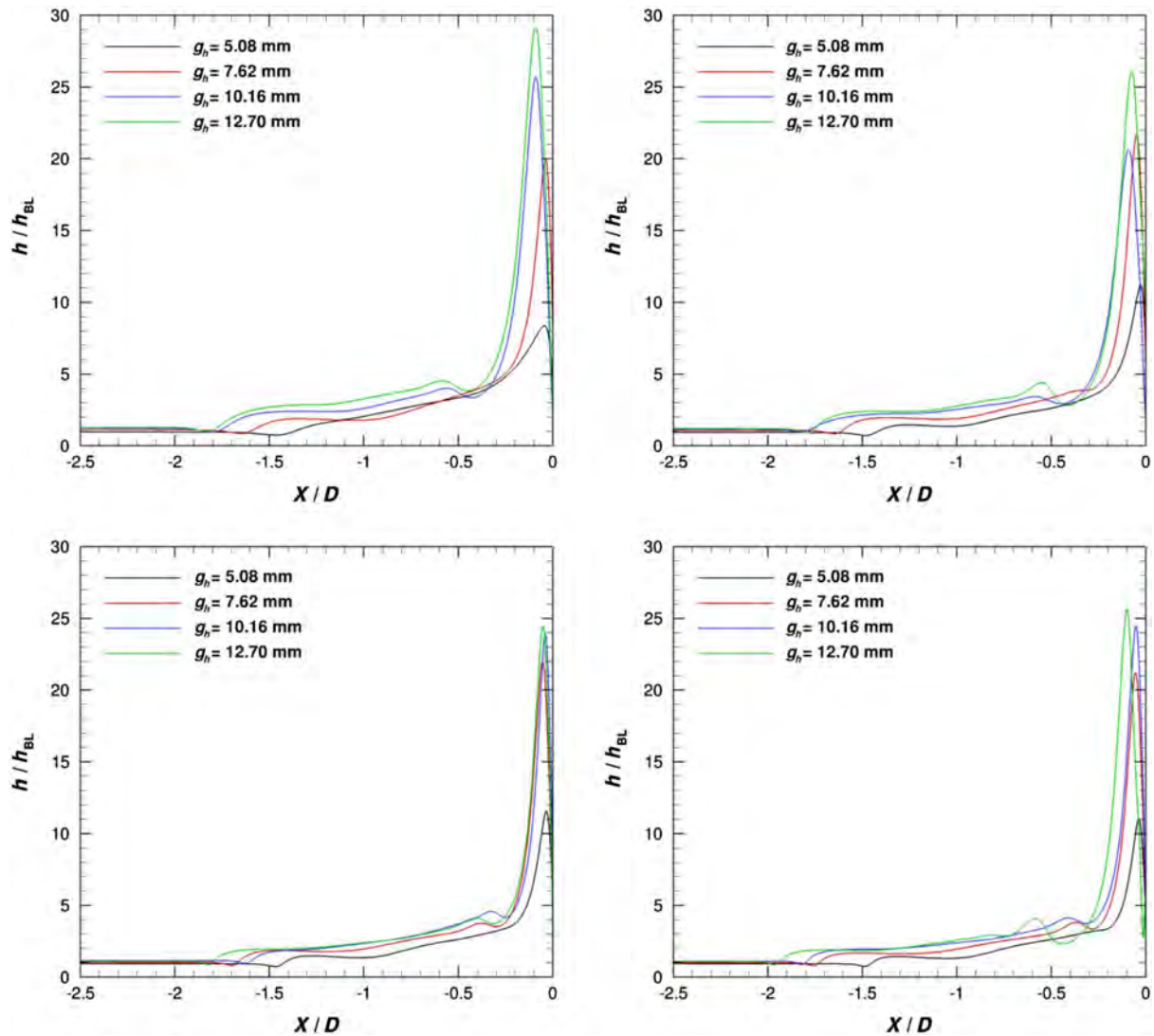


Fig. 22 Heat transfer coefficient normalized by baseline level on top centerline ($\phi = 0$ deg) in gap region. Angles of attack of 0, -4 , -8 , and -12 deg. Distance X measured from cylinder face.

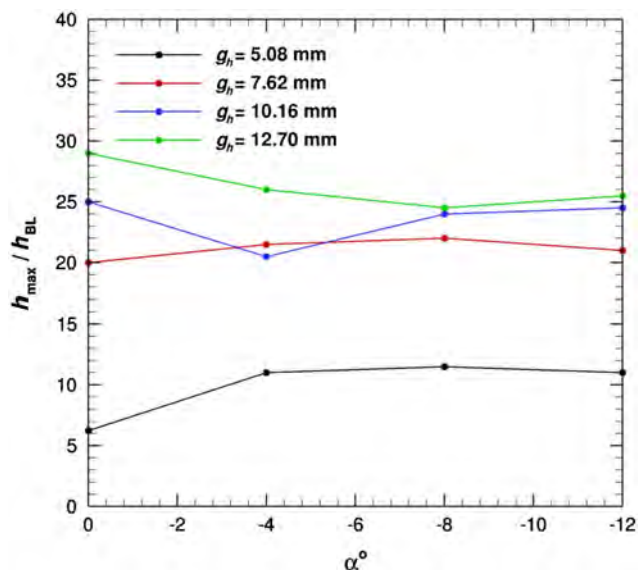


Fig. 23 Maximum normalized heat transfer coefficient on missile surface as a function of angle of attack.

The shape of the heat transfer on the cylinder surface differs between gap heights and angles of attack. As shown in Fig. 24, the smallest gap height, $g_h = 5.08$ mm, has maximum heating at the center of the symmetry line on the cylinder face at an angle of attack of 0 deg, whereas at larger angles of attack, such as at -12 deg, the maximum heating is located off-center. However, this may be nonphysical and due to the employed mesh; Atkinson et al. [18] showed that improved meshing techniques affected the location of maximum heat transfer on a hypersonic flow at angles of attack. Additionally, at differing gap heights, $g_h = 7.62$ mm and $g_h = 12.70$ mm, at the same angle of attack of -4 deg, the shape of the heating on the cylinder is not consistent. Collectively, there was no general trend in the qualitative shape of heat transfer on the cylinder surface.

Figure 25 plots the maximum heat transfer coefficient levels on the cylinder as a function of angle of attack. As expected, the heat transfer coefficient increases significantly as gap height increases, for all angles of attack. Interestingly, however, while heat transfer increases monotonically from $\alpha = 0$ to $\alpha = -8$, the behavior reverses from $\alpha = -8$ to $\alpha = -12$. In addition, the heat transfer levels were significantly above those computed on the missile surface, which ranged from 10 to 30 times the baseline level. Although the high rates of heat transfer on the cylinder surface only occur over a small surface area, they are severe enough to cause significant design concern in applications.

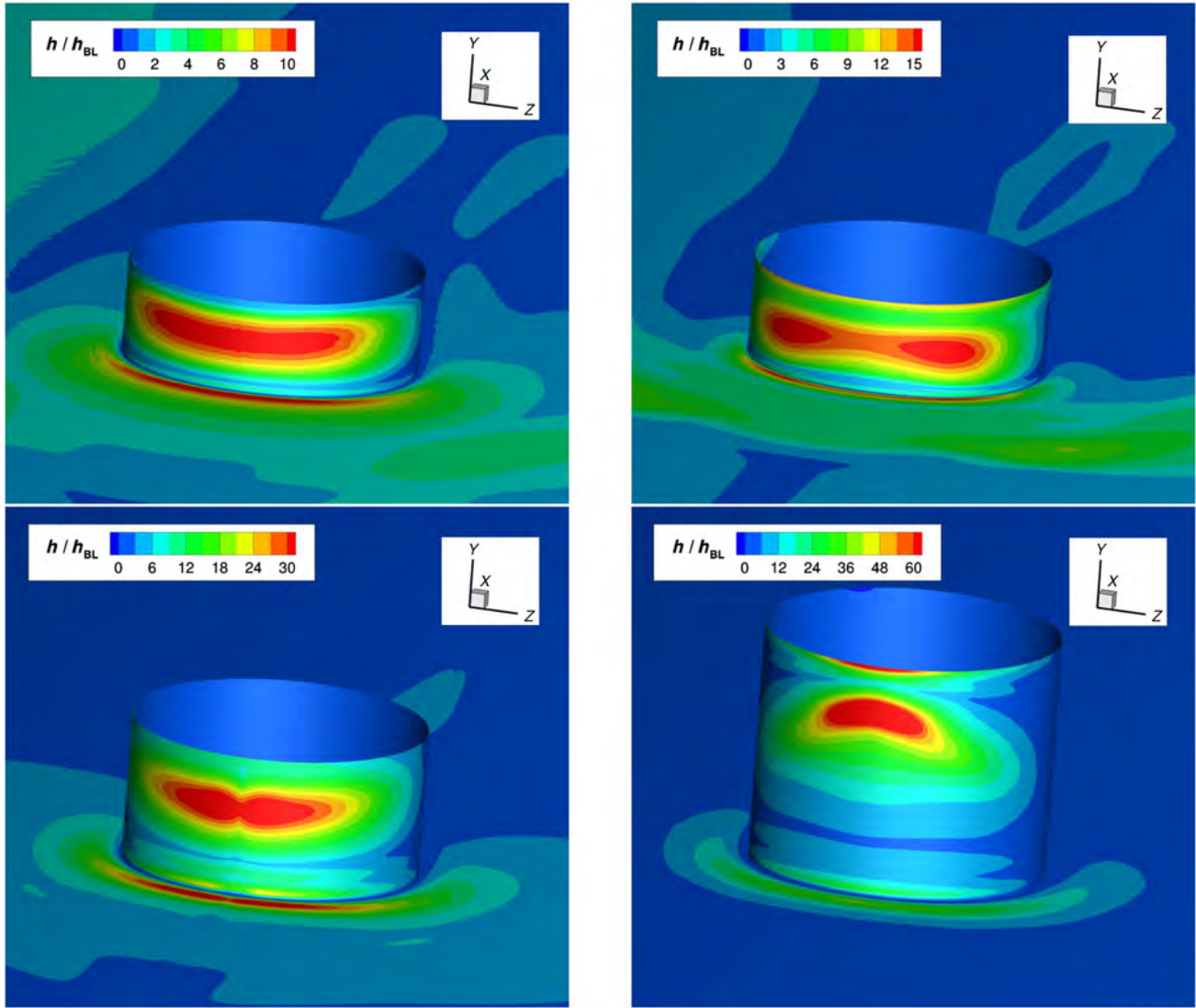


Fig. 24 Heating contour plot normalized by baseline level on missile and cylinder surfaces. Gap height of 5.08 mm at angles of attack of 0 and -12° (top), and gap heights of 7.62 and 12.70 mm at an angle of attack of -4° (bottom).

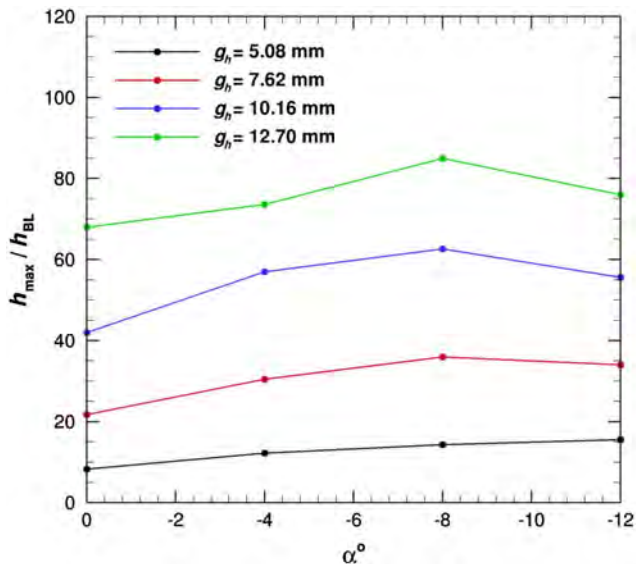


Fig. 25 Maximum normalized heat transfer coefficient on cylinder surface as a function of angle of attack.

V. Conclusions

This paper presents a computational investigation into high-speed flow over a missile with, and without, a fin and connecting cylinder. For each case, the computational mesh encompassed the full geometry; no symmetry boundary conditions were employed. The flow conditions, geometry, and parametric variations were chosen to match an experimental study conducted by Neumann and Hayes [12]. The primary focus of this work was to explore the aerothermodynamic loading, specifically aerodynamic heating, in the gap region of the missile-fin configuration.

The computed properties of the baseline configuration agreed reasonably well with experimental data. The predicted heat transfer coefficients and surface pressures on the windward side, for every computed angle of attack, fell within the uncertainty in the experimental measurements. Levels of these properties on the top centerline ($\phi = 0^\circ$) increased with increasing angle of attack, as observed experimentally. Computed pitot pressure profiles of the boundary layer and the shock wave over the top centerline ($\phi = 0^\circ$) also agreed well with experimental data. The computed boundary layer on the top centerline ($\phi = 0^\circ$) decreased in thickness with increasing angle of attack, as in the measurements.

Computations of the missile-fin configuration were organized in a parametric study that involved variations in angle of attack and the

gap height. Maximum levels of surface pressure and heat transfer on the missile surface upstream of the cylinder tended to increase with increasing gap height. Variations in angle of attack did not affect normalized maximum levels monotonically across gap heights. Skin friction and sectional streamline calculations supported the existence of flow separation in the gap region within two cylinder diameter lengths upstream of the cylinder surface. This interaction caused a spike in surface pressure, as large as 20 times the baseline level, and heat transfer, as large as 30 times the baseline level, near its reattachment point very close to the cylinder.

The maximum experimental values for heat transfer levels on the missile surface were never above 11 times the baseline level and always occurred at the closest probe to the cylinder, at $X/D = -0.3$. The maximum computed values occurred very close to the cylinder, a region that was inaccessible experimentally. In the computations, the heat transfer maxima were calculated to occur within 0.1 cylinder diameter length upstream of the cylinder surface. Maxima ranged from 10 times the baseline level, for $g_h = 5.08$ mm, to 30 times the baseline level, for $g_h = 12.70$ mm. Furthermore, computed heat transfer levels on the upstream side of the cylinder significantly exceeded those on the missile surface. The maximum heat transfer levels on the cylinder surface ranged from 10 to 80 times the baseline level. Heat transfer in this region was not measured in the wind tunnel tests.

Neumann and Hayes concluded that the region under the fin and upstream of the cylinder on the missile surface is the design point for the thermal protection of gapped control surfaces. The results presented in this paper support this conclusion and also suggest that the upstream side of the cylinder surface is another design point. The levels of heat transfer on the baseline missile surface are high for Mach 6 flow, and the heating near the missile-fin surface and cylinder is orders of magnitude larger. This produces large aerodynamic heating loads on the connecting cylinder and missile surface, which therefore need to be thermally protected.

Acknowledgments

The presented work was supported by the U.S. Air Force Research Laboratory under contract FA8650-18-C-2253, with program manager J. Boston. Computational resources were provided by the U.S. Air Force Research Laboratory high-performance computing cluster Mustang. The authors would also like to acknowledge collaborators from the University of Tennessee, Knoxville, Ryan Glasby and Ethan Hereth, who helped facilitate the research for this project. This paper, and the contained material, was assigned a clearance of CLEARED (case number 2021-0244) on 13 May 2021.

References

- [1] Gudmundsson, S., *General Aviation Aircraft Design Applied Methods and Procedures*, 1st ed., Butterworth-Heinemann, Oxford, England, U.K., 2014, pp. 714–740.

- [2] Déler, J., Marvin, J. G., and Reshotko, E., “Shock-Wave Boundary Layer Interactions,” Advisory Group for Aerospace Research and Development TR 280, Neuilly-Sur-Seine, France, 1986.
- [3] Gaitonde, D. V., “Progress in Shock Wave/Boundary Layer Interactions,” *Progress in Aerospace Sciences*, Vol. 72, Jan. 2015, pp. 80–99. <https://doi.org/10.1016/j.paerosci.2014.09.002>
- [4] Tauber, M. E., “A Review of High-Speed, Convective, Heat-Transfer Computation Methods,” NASA Office of Management TR 2914, 1989.
- [5] Stern, I., and Rowe, W. H., “Effect of Gap Size on Pressure and Heating over the Flap of a Blunt Delta Wing in Hypersonic Flow,” *Journal of Spacecraft and Rockets*, Vol. 4, No. 1, 1967, pp. 109–114. <https://doi.org/10.2514/3.28818>
- [6] Jing, X., Haiming, H., Guo, H., and Song, M., “Numerical Simulation of Supersonic Gap Flow,” *PLoS One*, Vol. 10, No. 1, Jan. 2015, pp. 1–16. <https://doi.org/10.1371/journal.pone.0117012>
- [7] Cooper, L., and Putz, K. E., “Generalized Flow in Gaps and Slots Including the Effects of Ablation,” *Journal of Spacecraft and Rockets*, Vol. 11, No. 5, 1974, pp. 287–294. <https://doi.org/10.2514/3.62065>
- [8] Mikhail, A. G., “Assessment of Two Fast Aerodynamic Codes for Guided Projectiles,” *Journal of Spacecraft and Rockets*, Vol. 24, No. 4, 1987, pp. 303–310. <https://doi.org/10.2514/3.25917>
- [9] Hunt, L. R., “Aerothermal Environment in Chordwise Gaps Between Split Elevons at Mach 6.8,” NASA TR 1783, 1980.
- [10] Wong, H., and Kremer, F., “Numerical Assessment on the Heating of the Rudder/Fin Gap in X38 Space Vehicle,” *Proceedings of the Third European Symposium on Aerothermodynamics for Space Vehicles*, ESA Special Publication, Vol. 426, Noordwijk, The Netherlands, 1999, pp. 77–85.
- [11] Bertin, J., and Goodrich, W., “Aerodynamic Heating for Gaps in Laminar and Transitional Boundary Layers,” *18th Aerospace Sciences Meeting*, AIAA Paper 1980-0287, 1980. <https://doi.org/10.2514/6.1980-287>
- [12] Neumann, R. D., and Hayes, J. R., “Aerodynamic Heating in the Fin Interaction Region of Generalized Missile Shapes at Mach 6,” Air Force Flight Dynamics Lab. TR 79-3066, Wright-Patterson AFB, OH, 1979.
- [13] Sorenson, R. L., “The 3DGRAPE Book: Theory, Users’ Manual, Examples,” NASA TM-102224, 1989.
- [14] McDaniel, D. R., Tuckey, T. R., and Morton, S. A., “The HPCMP CREATETM-AV Kestrel Computational Environment and its Relation to NASA’s CFD Vision 2030,” *55th AIAA Aerospace Sciences Meeting*, AIAA Paper 2017-0813, 2017. <https://doi.org/10.2514/6.2017-0813>
- [15] Cengel, Y., and Ghajar, A., *Heat and Mass Transfer Fundamentals and Applications*, 5th ed., McGraw-Hill, New York, 2014, p. 26.
- [16] van Driest, E. R., “The Problem of Aerodynamic Heating,” *Aeronautical Engineering Review*, Vol. 15, No. 10, Oct. 1956, pp. 26–41.
- [17] Gruenewald, K. H., “Temperature Recovery Factors in the Transitional and Turbulent Boundary Layer on a 40-Degree Cone Cylinder at Mach Number 2.9,” Naval Ordnance Lab. TR 2742, White Oak, MD, 1953.
- [18] Atkinson, M., Poggie, J., and Camberos, J., “Hypersonic Flow Computations for an Elliptic Cone at High Angle of Incidence,” *Journal of Spacecraft and Rockets*, Vol. 49, No. 3, 2012, pp. 496–506. <https://doi.org/10.2514/1.A32062>

S. Silton
Associate Editor